

Ch 12: Surface Areas and Volumes

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Two cubes each of volume 64 cm^3 are joined end to end.

Find the surface area of the resulting cuboid.

Q2. A cylindrical pencil sharpened at one end is a combination of which two solids? Name both.

Q3. A surahi is a combination of which two solids? Identify both from the shape.

Q4. A shuttle cock used in badminton is a combination of which two solids?

Q5. A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each end. The length of the entire capsule is 14 mm and the diameter is 5 mm. Find the total surface area.

Q6. Find the total surface area of a hemisphere of radius 7 cm.

Q7. If the total surface area of a solid hemisphere is 462 cm^2 , find its radius.

Q8. The radius of a solid sphere is 6 cm. Find its surface area and volume.

Q9. A cone of height 16 cm and base radius 12 cm is placed on a cylinder of height 10 cm and same base radius. Without finding the total surface area, write which surfaces are visible and which are hidden.

Q10. A cylinder and a cone have equal base radii and equal heights. Write the ratio of their volumes.

Q11. A cylinder, a cone, and a hemisphere have the same base radius r and same height h . Write the ratio of their volumes.

Q12. Find the volume of the largest right circular cone that can be cut out of a cube of side 7 cm.

Q13. Find the volume of the largest sphere that can be carved out of a solid cube of side 7 cm.

Q14. A solid metallic sphere of radius 4.2 cm is melted and recast into a cylinder of radius 6 cm. Find the height of the cylinder.

Q15. A cone of height 24 cm and base radius 6 cm is reshaped into a sphere. Find the radius of the sphere.

Q16. A metallic cuboid of dimensions $9\text{ cm} \times 8\text{ cm} \times 2\text{ cm}$ is melted into a cube. Find the side of the cube.

Q17. Find the number of solid spheres of radius 3 cm that can be made from a solid cylinder of radius 6 cm and height 12 cm.

Q18. A solid sphere of diameter 6 cm is dropped into a cylinder of radius 6 cm containing water. Find the rise in water level.

Q19. Two solid cones have the same volume. Their heights are in ratio 1:4. Find the ratio of their base radii.

Q20. If the radius of a sphere is doubled, by how many times does its volume increase?

Q21. If the height of a cylinder is halved, by how many times does its volume decrease?

Q22. A hemispherical depression of radius 7 cm is cut out from one face of a cube of side 14 cm. Find the total surface area of the remaining solid.

Q23. A solid is formed by joining a hemisphere of radius r to the base of a cone of height $2r$ and base radius r . Write the total height of this solid.

Q24. Water flows through a cylindrical pipe of diameter 2 cm at 3 m/s. Find the volume of water discharged per second.

Q25. A solid right circular cylinder of radius r cm and height h cm just encloses a sphere. What is the radius of the sphere?

Q26. The diameter of a metallic ball is 4.2 cm and the density of metal is 8.9 g/cm^3 . Find the mass of the ball.

Q27. 64 small spheres are made by melting one large sphere of radius 8 cm. Find the radius of each small sphere.

Q28. A copper rod of diameter 1 cm and length 8 cm is drawn into a wire of length 18 m. Find the thickness of the wire.

Q29. Find the number of coins of diameter 1.75 cm and thickness 2 mm that must be melted to form a cuboid of dimensions $5.5 \text{ cm} \times 10 \text{ cm} \times 3.5 \text{ cm}$.

Q30. Two cones of equal height and base radii 6 cm and 8 cm are placed base to base. Write the total surface area of the combined solid.

Q31. A solid hemisphere of radius r is placed on a cylinder of same radius and height r . Is the total surface area $= 5\pi r^2$? Verify.

Q32. A right circular cone has the same curved surface area as a cylinder of equal base radius. If the slant height of the cone equals the height of the cylinder, find the ratio of their curved surface areas directly.

Q33. A toy is a combination of a hemisphere of radius 3.5 cm and a cone of same base radius and height 4 cm. Find the volume of the toy.

SECTION B - 3 MARKS

Q1. A solid is in the form of a cylinder with hemispherical ends. The total height is 19 cm and the diameter is 7 cm. Find the volume and total surface area of the solid. (PYQ 2019)

Q2. From a solid cylinder of height 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid.

Q3. A wooden toy rocket is in the shape of a cone mounted on a cylinder. The height of the cone is 16 cm, height of the cylinder is 48 cm, and the base radius of both is 6 cm. Find the volume to be painted in each colour if the cone is painted orange and the cylinder yellow.

Q4. A vessel is in the form of an inverted cone of height 8 cm and radius 5 cm. It is filled with water to the brim. When spherical lead shots of radius 0.5 cm are dropped in, one-fourth of water flows out. Find the number of lead shots dropped.

Q5. A heap of rice is in the form of a cone of base diameter 24 m and height 3.5 m. Find the volume of rice. How much canvas cloth is required to just cover the heap?

Q6. A tent is in the shape of a cylinder of height 2.1 m and diameter 4 m, surmounted by a cone of slant height 2.8 m. Find the area of canvas needed and the cost at Rs. 500 per m^2 . (PYQ 2019)

Q7. An ice cream cone consists of a cone of height 12 cm and diameter 7 cm with a hemispherical scoop of ice cream on top of diameter 7 cm. Find the volume of the ice cream.

Q8. A gulab jamun is shaped like a cylinder with two hemispherical ends. The total length is 5 cm and diameter is 2.8 cm. Find the volume of one gulab jamun. If 45 such gulab

jamuns are made and 30% of the volume is sugar syrup, find the amount of syrup used.

Q9. A solid cylinder of base radius 3 cm and height 5 cm is melted to form cones each of radius 1 cm and height 1 cm. Find the number of cones. (PYQ 2020)

Q10. A metallic solid sphere of radius 10.5 cm is melted and recast into smaller solid cones of radius 3.5 cm and height 3 cm. Find the number of cones formed. (PYQ 2020)

Q11. A 20 m deep well of diameter 7 m is dug and the earth taken out is spread evenly to form a platform $22 \text{ m} \times 14 \text{ m}$. Find the height of the platform.

Q12. A well of diameter 3 m is dug 14 m deep. The earth is spread evenly all around it in a ring of width 4 m to form an embankment. Find the height of the embankment.

Q13. Three metallic solid cubes of edges 3 cm, 4 cm, and 5 cm are melted into a single cube. Find the edge of the cube.

Q14. How many spherical lead shots of diameter 4 cm can be made out of a solid cube of lead of edge 44 cm?

Q15. A container shaped like a right circular cylinder of diameter 12 cm and height 15 cm is full of ice cream. Ice cream is to be filled into cones of height 12 cm and diameter 6 cm with a hemispherical top. Find the number of cones that can be filled.

Q16. Water in a canal 6 m wide and 1.5 m deep flows at 10 km/hr. How much area will it irrigate in 30 minutes if 8 cm of standing water is needed?

Q17. A farmer connects a pipe of internal diameter 20 cm from a canal into a cylindrical tank of diameter 10 m and depth 2 m. If water flows at 3 km/hr, in how many hours will the tank be filled?

Q18. A solid metallic hemisphere of radius 8 cm is melted and recast into a right circular cone of base radius 6 cm. Find the height of the cone. (PYQ 2020)

Q19. The largest possible sphere is carved out of a wooden cube of side 7 cm. Find the volume of wood wasted.

Q20. A container is in the shape of an inverted cone of height 8 cm and base radius 5 cm. It is filled with water. Spheres of radius 1 cm are dropped in and two-fifths of water overflows. Find the number of spheres dropped. (PYQ 2022)

Q21. A pen stand is a cuboid of dimensions $15\text{ cm} \times 10\text{ cm} \times 3.5\text{ cm}$ with 4 conical depressions of radius 0.5 cm and depth 1.4 cm. Find the volume of wood in the pen stand.

Q22. Selvi's house has a cylindrical overhead tank of radius 60 cm and height 95 cm filled by a sump (cuboid) of dimensions $1.57\text{ m} \times 1.44\text{ m} \times 95\text{ cm}$. After filling the tank, find the height of water left in the sump.

Q23. Water flows at 10 m per minute through a cylindrical pipe of diameter 5 mm. How long will it take to fill a conical vessel of diameter 40 cm and depth 24 cm?

Q24. A solid cylinder of radius 6 cm and height 12 cm is melted and 3 smaller cylinders of radius 4 cm are formed. Find the height of each smaller cylinder.

Q25. A solid hemisphere of radius 12 cm and a solid cone of base radius 12 cm and height 24 cm are melted together. The mixture is poured into a cylinder of radius 12 cm. Find the height of the mixture in the cylinder.

Q26. A juice seller serves customers using cylindrical glasses of inner diameter 5 cm and height 10 cm. The bottom of each glass has a hemispherical raised portion of radius 2.5 cm. Find the actual capacity of the glass. (Use $\pi = 3.14$)

Q27. 504 cones each of diameter 3.5 cm and height 3 cm are melted and recast into a metallic sphere. Find the diameter of the sphere. (PYQ 2020)

Q28. A rain harvesting system collects rain from a roof $22 \text{ m} \times 20 \text{ m}$ into a cylindrical tank of diameter 2 m and height 3.5 m. If the tank is just full, find the rainfall in cm.

Q29. A solid is in the form of a cone of height 4 cm standing on a hemisphere of radius 4 cm. A cylinder of radius 8 cm and height 12 cm just surrounds this solid. Find the volume of water that can fill the space between the solid and the cylinder.

Q30. A toy manufacturer makes a toy shaped like a cylinder with two conical ends. The total length is 18 cm, diameter is 4 cm, and each cone is 3 cm long. Find the total surface area of the toy.

SECTION C - 5 MARKS

Q1. A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. Find the volume of the toy. If a right circular cylinder just circumscribes the toy, find how much more space does the cylinder have than the toy.

Q2. A solid iron pole consists of a cylinder of height 220 cm and base radius 24 cm surmounted by another cylinder of height 60 cm and radius 8 cm. Find the mass of the pole if density of iron is 8 g/cm^3 . (PYQ 2019)

Q3. From a solid cylinder of height 7 cm and base radius 12 cm, a conical cavity of the same height and same base is hollowed out. Find the volume and total surface area of the remaining solid.

Q4. A rocket is in the form of a cylinder closed at the lower end and surmounted by a cone of the same radius. The diameter of the cylinder is 6 cm and height is 12 cm. The slant height of the

cone is 5 cm. Find the total surface area and volume of the rocket. (PYQ 2020)

Q5. Due to heavy floods, 100 tents are needed immediately. Each tent is cylindrical with diameter 4.2 m and height 3 m, surmounted by a conical top of height 2.8 m. Find the cost of 100 tents at Rs. 100 per m^2 of canvas.

Q6. A vessel in the form of a hemisphere is surmounted by a hollow cylinder of same radius. The total height of the vessel is 13 cm and the diameter is 14 cm. Find the inner surface area. Also find the volume of water it can hold.

Q7. A hemispherical tank full of water is emptied by a pipe at $25/7$ litres per second. The tank has a diameter of 3 m. How much time will it take to empty the tank?

Q8. An oil funnel is a combination of a cylinder of height 2 cm and diameter 4 cm and a cone of slant height 12 cm and radius same as the cylinder. Find the total surface area of the funnel and the volume of oil it can hold.

Q9. Case Study: A school designs a model trophy in the shape of a cylinder surmounted by a hemisphere. The diameter of both is 7 cm. The cylinder's height is 13.5 cm. Find: (i) the curved surface area of the cylinder (ii) the curved surface area of the hemisphere (iii) the total surface area of the trophy (iv) the volume of the trophy.

Q10. A solid consisting of a right circular cone of height 120 cm and radius 60 cm standing on a hemisphere of radius 60 cm is

placed in a cylinder full of water such that it touches the bottom. If the cylinder's radius is 60 cm and height is 180 cm, find the volume of water left in the cylinder.

Q11. Water flows at 10 km/hr through a pipe of internal diameter 7 cm into a rectangular tank of dimensions 50 m $\times$ 44 m. Find the time for the level to rise by 21 cm.

Q12. A well of diameter 14 m is dug 15 m deep. The earth dug out is spread to form a platform 44 m long and 27 m wide. Find the height of the platform.

Q13. Case Study: A municipality collects rainwater. A cylindrical tank of radius 60 cm and height 2 m is kept on the roof. Rainwater flows from a square roof of side 10 m. Find: (i) the volume of the tank (ii) the amount of rainfall in mm needed to completely fill the tank (iii) if rainfall is 2 cm, find the volume of water collected on the roof (iv) the number of such full tanks this volume can fill.

Q14. A bucket is in the shape of a cylinder of height 28 cm with a hemispherical depression at the bottom. The base radius is 7 cm. Find the inner surface area of the bucket and the volume of water it can hold.

Q15. Milk in a cylindrical container of radius 30 cm and height 40 cm is transferred to two containers simultaneously — one conical of radius 10 cm and height 18 cm, and one spherical of radius 6 cm. After filling both, find the height of milk remaining in the cylindrical container.

Q16. A solid iron pole consists of a cylinder of height 1.8 m and base diameter 28 cm. On the top is a hemisphere of same radius. Find the volume of iron used and the mass if density is 8 g/cm^3 .

(PYQ)

Q17. The interior of a building is in the form of a right circular cylinder of diameter 4.3 m and height 3.8 m surmounted by a cone whose vertical angle is 90° . Find the area of the surface, the floor is excluded, to be painted at the rate of Rs. 25 per m^2 .

Q18. A spherical ball of diameter 21 cm is melted and recasted into small cones of height 3 cm and base radius 1.5 cm. Find the number of cones obtained and the ratio of the surface area of the sphere to the total surface area of all the cones.

Q19. Case Study: A toy factory makes toys of the following description — a solid hemisphere of radius 3.5 cm is placed on a cylindrical base of radius 3.5 cm and height 5 cm with a conical hat of radius 3.5 cm and height 4 cm on top. Find: (i) the total height of the toy (ii) the volume of each part (iii) the total volume (iv) the total surface area.

Q20. A well of diameter 4 m and depth 21 m is dug. The earth taken out is spread uniformly to form a circular ring-shaped embankment of inner diameter 4 m and width 3 m. Find the height of the embankment.

Q21. A school supplies 120 students with cylindrical pencils of length 25 cm and circumference 1.5 cm. The pencils are coloured on the curved surface at Rs. 0.05 per dm^2 . Find the total cost of colouring all pencils.

Q22. A solid is composed of a cylinder of height 12 cm and radius 6 cm with a hemisphere of radius 6 cm attached at the top and a cone of radius 6 cm and height 8 cm attached at the bottom. Find the total surface area and volume of the composite solid.

Q23. An overhead water tank in the form of a cylinder of radius 70 cm and height 1.5 m is to be painted on the outside including the top and bottom. Find the cost of painting at Rs. 15 per 100 cm^2 .

Q24. In a hospital, a patient is given medicine in a cylindrical container of radius 1 cm and height 5 cm. He empties it into a conical vessel of radius 2 cm and height x cm. If the conical vessel is exactly full, find x . What does this tell us about the relationship between volumes of the two vessels?

Q25. Case Study: An architect designs a decorative pillar consisting of four parts — a square base of side 20 cm and height 10 cm, a cylinder of radius 7 cm and height 60 cm, a hemisphere of radius 7 cm on top, and an inverted cone with base radius 7 cm and height 10 cm at the very top for aesthetics but solid. Find: (i) the volume of each part (ii) the total volume (iii) if the entire pillar is made of marble at Rs. 5 per cm^3 , find the total cost.